EE6301
Reviewed Quiz Guide
概念覆盖复核版

25 MCQs · Concepts and short calculations
25道单选 · 概念与短计算

Body text / 正文：11 pt
English first, Chinese immediately below. / 英文在前，中文紧随。

2026-10-11 · Course notes, sample explanations and original practice
课件知识、样题参考解析与原创练习

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Quiz scope and format / Quiz范围与形式

EN: Use this edition first: it reselects Quiz priorities using eight slide decks, eight transcripts, the Spring sample and past final materials. English and Chinese correspond throughout. The 25-question paper is original practice.

中文： 这次Quiz先用本版：结合八份课件、八份字幕、春季样题与历年期末材料重筛重点，全文中英文对应。25题卷为原创模拟练习。

EN: Confirmed format: 25 single-answer multiple-choice questions, 4 marks each, 60 minutes, closed book. The lecturer later corrects an earlier reference to 35 questions to 25. The A4 double-sided sheet is for the final exam, not this Quiz. Scope: Weeks 1–7 and Week 8 Parts One and Two, physical PDF p5–29, ending at filters before Human–Machine Interface.

中文： 确认形式：25道单选，每题4分，60分钟，闭卷。老师后段将先前的35题更正为25题；A4双面纸用于期末，不用于这次Quiz。范围为Week1–7与Week8前两章（实际PDF p5–29），到滤波器为止，Human–Machine Interface之前。

EN: The lecturer estimates 60–70% concepts and 30–40% very short calculations. The practice paper therefore uses 17 concept questions and 8 calculations (68%/32%); this is a training allocation, not a promise about the exact distribution of the real paper. Memorize key terms and short relations. The lecturer’ s informal “short equation” description is not a literal character-count rule.

中文： 老师估计60–70%概念、30–40%很短的计算。模拟卷据此安排17道概念、8道计算（68%/32%）；这是训练配比，不承诺真卷恰好如此。要记核心英文名词与短关系式；老师口头形容“短公式”，不能机械按字符个数判断。

EN: Scope and format anchors: T8-0035–0036, T8-0060–0064. Transcript IDs are 150-word text segments created for retrieval, not timestamps. The source TXT files contain no timestamps. Slide pages always mean physical PDF page order.

中文： 范围与形式依据：T8-0035–0036、T8-0060–0064。字幕段号为便于定位而创建的150词文本段，不是时间戳；原TXT没有时间戳。课件页码一律指实际PDF页序。

What to study first / 先学什么

EN: A: direct Quiz promises or sample counterparts. Start with LOD, ART/LSPR, objective NA comparisons, grating density, fluorescence/quantum yield, Beer–Lambert, OCT/CCD distinctions, and patch clamp/MEA. B: repeated foundations that connect these questions, including sensor structure, basic electrochemistry, action-potential phases, differential amplification and filter roles.

中文： A：老师明确提到Quiz或已有样题同类。先学LOD、ART/LSPR、物镜NA比较、光栅密度、荧光与量子产率、Beer–Lambert、OCT/CCD辨析、patch clamp与MEA。B：连接这些题目的反复讲解基础，包括传感器结构、基础电化学、动作电位阶段、差分放大与滤波作用。

EN: C: secondary or mainly final-exam calculations, such as long OCT intensity expressions, complex impedance/Bode work, detailed ultrasound resolution, and long SPR or multi-ion derivations. D: explicitly excluded local details or beyond-scope material, including Week 8 p30 onward and conformal-electrode technical design. These categories are evidence-based study priorities, not numerical probabilities.

中文： C：次要或主要用于期末计算，如OCT长强度式、复杂阻抗/Bode、详细超声分辨率、长SPR或多离子推导。D：明确局部排除或超范围内容，包括Week8 p30起与贴合电极技术设计。分类表示依据与复习顺序，不表示数字化出题概率。

EN: Important correction: sample Q10 is B, 5 viruses, following the lecturer’ s explicit integer-threshold explanation. The earlier A≈4 answer was wrong. Continuous concentration LOD values are not rounded this way.

中文： 重要纠错：样题Q10依据老师明确的整数阈值解释，答案改为B：5个病毒。之前A≈4的答案错误。连续浓度LOD不按这种方式取整。

EN: Coverage safeguard: the lecturer says the sample questions are only examples (T8-0060). Study all in-scope concepts, not only predicted questions. This reviewed edition keeps 25 main cards and adds 32 concise recognition entries. A/B set study order; C lowers calculation depth, not all related concepts.

中文： 覆盖提醒：老师明确说样题只是示例（T8-0060）。范围内概念都应复习，不能只背预测题。本复核版保留25张主卡，补全32条简短辨识知识。A/B决定复习顺序；C主要降低计算深度，不是把相关概念全部删去。

Read the English question correctly / 先看懂英文题目

EN: Which statement is incorrect / wrong?

> **中文：** 哪项陈述错误？优先找错误方向或条件。

EN: Which is NOT related to…?

> **中文：** 哪项不属于……？不要漏看NOT。

EN: Minimum detectable number

> **中文：** 最小可检测数量；若是整数个体，需达到连续阈值。

EN: Linear region / calibration slope

> **中文：** 线性区 / 校准斜率；排除饱和点。

EN: Assume all other conditions are identical

> **中文：** 假设其他条件相同；才能只比较一个因素。

EN: Higher resolving power / smaller resolvable separation

> **中文：** 更高分辨本领 / 更小可分辨间距：都表示分辨更好。

EN: Incident / absorbed / emitted photons

> **中文：** 入射 / 吸收 / 发射光子：量子产率分母用吸收光子。

EN: Reference electrode / reference arm

> **中文：** 参比电极 / 参考光臂：一个用于电位基准，一个用于干涉比较。

EN: Intracellular / extracellular

> **中文：** 细胞内 / 细胞外：膜电位定义的相减方向不能反。

EN: Round trip / one way

> **中文：** 往返 / 单程：决定传播时间算深度是否除以2。

EN: Common-mode / differential-mode

> **中文：** 共模 / 差模：两输入共有分量 / 两输入差值。

EN: High-pass / low-pass / band-pass / notch

> **中文：** 高通 / 低通 / 带通 / 陷波：按保留或抑制的频率判断。

A and B focus cards / A与B重点卡片

01. [B] Sensor, transducer and biosensor / 传感器、换能器与生物传感器

EN: A sensor responds to a physical or chemical input. A transducer converts a recognition event or another input into a measurable signal. A biosensor combines a biological recognition element with a transducer; the recognition element gives specificity, while the transducer produces the readout.

中文： Sensor响应物理或化学输入；transducer把识别事件或其他输入转为可测信号。Biosensor由生物识别元件与换能器组成：识别元件赋予特异性，换能器产生读出信号。

EN: Watch the wording: Do not confuse the analyte (the target being measured), the bioreceptor (what recognizes it), and the transducer (what converts recognition into a signal).

中文： 读题易错点：不要混淆analyte（被测目标）、bioreceptor（识别目标的元件）和transducer（把识别转成信号的元件）。

Source / 来源：W1 PDF p13–20; W1 PDF p39–42; W2 PDF p6–12 | T1-0017–0021, T1-0033–0034, T2-0004–0009

02. [A] Sensitivity, selectivity and limit of detection / 灵敏度、选择性与检测限

EN: For a linear calibration $y=mC+b$, sensitivity is the slope m. Selectivity means discriminating the target from interfering species. Under the course convention, $LOD=3\sigma/|m|$, where σ is the standard deviation of the background response. Holding the other factor fixed, a larger absolute slope or smaller background noise lowers the LOD.

中文： 线性校准$y=mC+b$中，灵敏度是斜率m。选择性是区分目标与干扰物的能力。本课程采用$LOD=3\sigma/|m|$，σ是背景响应的标准差。在另一因素固定时，斜率绝对值更大或背景噪声更小，才会降低检测限。

EN: Watch the wording: Use the linear region rather than a saturated endpoint. A low LOD means better detection capability. Diagnostic sensitivity TP/(TP+FN) is a different meaning of sensitivity; do not substitute it for calibration slope.

中文： 读题易错点：用线性区计算，不能把饱和点纳入斜率。LOD越小，检测能力越好。诊断学sensitivity=TP/(TP+FN)是另一个含义，不能代替校准斜率。

Source / 来源：W1 PDF p58–61; W1 PDF p60–62; W6 p49 | T1-0053–0058, T6-0050–0053

03. [A] Integer detection threshold: corrected sample Q10 / 整数检测阈值：修正样题Q10

EN: In the sample graph, the linear interval is 20–80 viruses: $m=(700-100)/(80-20)=10$ pF/virus. With $\sigma=14$ pF, LOD=4.2 viruses. The lecturer explicitly rejects 4 and chooses the minimum integer count that reaches the threshold: 5, option B.

中文： 样题图线性区为20–80个病毒：$m=(700-100)/(80-20)=10$ pF/virus。$\sigma=14$ pF时，LOD=4.2个病毒。老师明确否定4，并选择达到阈值的最小整数数量：5，对应B。

EN: Watch the wording: The previous guide incorrectly used the nearest option, A=4. This edition corrects it to B=5. Round up only when the question asks for a minimum detectable integer count; a concentration LOD such as 0.42 μM stays continuous.

中文： 读题易错点：旧版错误地选了最近选项A=4，本版改为B=5。只有题目要求“最小可检测整数数量”时才向上取整；0.42 μM这样的浓度检测限仍保留连续数值。

Source / 来源：Sample quiz p2 Q10; W1 p59–62 | T7-0057–0060

04. [A] Absorbance and Beer–Lambert law / 吸光度与比尔—朗伯定律

EN: Absorbance is $A=-\log_{10}(I/I_0)=\varepsilon CL$. Transmittance is $T=I/I_0$, so $A=-\log_{10}T$. If C is in mol/L and L in cm, ε must be in L mol^{-1} cm^{-1}. A=1 means 10% transmitted light; A=2 means 1%.

中文： 吸光度$A=-\log_{10}(I/I_0)=\varepsilon CL$；透射率$T=I/I_0$，所以$A=-\log_{10}T$。若C用mol/L、L用cm，ε必须用L mol^{-1} cm^{-1}。A=1代表透过10%，A=2代表透过1%。

EN: Watch the wording: The lecturer names A=εCL as a short equation to remember for the closed-book Quiz. Absorbance is dimensionless and is not the same as an absorbed fraction or a current.

中文： 读题易错点：老师明确举A=εCL为闭卷Quiz应记忆的短公式。Absorbance无量纲，不等于吸收比例，也不是电流。

Source / 来源：W2 PDF p36–40; W2 PDF p41–43; Sample quiz p2 Q11; Sample quiz p3 Q12; W2 p37–43; W1 p61–62 | T2-0041–0051, T7-0060–0061

05. [A] Fluorescence and Stokes shift / 荧光与斯托克斯位移

EN: Fluorescence follows absorption and relaxation in an excited state. The usual Stokes shift gives a longer emission wavelength and lower photon energy than excitation. An excitation filter selects excitation light; an emission filter and dichroic mirror help separate fluorescence from the excitation beam.

中文： 荧光经历吸收及激发态弛豫。通常的Stokes shift使发射波长比激发波长更长、光子能量更低。Excitation filter选择激发光，emission filter与dichroic mirror帮助分离荧光和激发光。

EN: Watch the wording: If a question asks for the wrong statement, check the direction of wavelength and energy carefully. Two-photon excitation combines two lower-energy photons; it does not mean two individually higher-energy photons.

中文： 读题易错点：若题目问wrong statement，先核对波长与能量的方向。Two-photon excitation是两个低能光子共同激发，不是两个各自更高能的光子。

Source / 来源：W2 PDF p69–76; W2 PDF p77–79; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61 | T2-0088–0100, T7-0051–0053

06. [A] Quantum yield, lifetime and radiative rate / 量子产率、寿命与辐射速率

EN: Quantum yield Φ is emitted photons divided by absorbed photons. With total non-radiative rate k_{nr}, $\tau=1/(k_r+k_{nr})$, $\Phi=k_r\tau$ and $k_r=\Phi/\tau$. Fluorescence brightness also depends on absorption, commonly through εΦ in the course examples.

中文： 量子产率Φ是发射光子数除以吸收光子数。非辐射总速率为k_{nr}时，$\tau=1/(k_r+k_{nr})$、$\Phi=k_r\tau$、$k_r=\Phi/\tau$。本课程示例中亮度还受吸收影响，常用εΦ表示。

EN: Watch the wording: Do not use all incident photons as the denominator unless all are absorbed. Lifetime is an exponential 1/e decay time, not automatically a half-life. Convert ns and s consistently.

中文： 读题易错点：除非入射光子全部被吸收，否则分母不能用全部入射光子。Lifetime是指数衰减至1/e的时间，并非直接等于半衰期。ns与s必须统一。

Source / 来源：W2 PDF p82–88; Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T2-0105–0107, T7-0054–0055

07. [A] Raman, Rayleigh, Stokes and anti-Stokes / 拉曼、瑞利、斯托克斯与反斯托克斯

EN: Rayleigh scattering is elastic: scattered photon energy is unchanged. Raman scattering is inelastic and reports molecular vibrations. Stokes Raman light has lower energy and longer wavelength; anti-Stokes has higher energy and shorter wavelength. Raman shift is usually expressed in cm^{-1}.

中文： Rayleigh scattering是弹性散射，光子能量不变。Raman scattering是非弹性散射，反映分子振动。Stokes Raman能量更低、波长更长；anti-Stokes能量更高、波长更短。拉曼位移常用cm^{-1}。

EN: Watch the wording: Fluorescence and Raman are different mechanisms. In the sample resonator question, Raman is the clearest non-resonance-sensing option. SERS means surface-enhanced Raman scattering/spectroscopy; the metal nanostructure enhances the Raman signal.

中文： 读题易错点：荧光和拉曼机制不同。样题的共振器辨析中，Raman是最明确的非共振传感选项。SERS是surface-enhanced Raman scattering/spectroscopy，金属纳米结构增强拉曼信号。

Source / 来源：W2 PDF p55–62; W2 PDF p65–68; W3 p88–94 | T2-0070–0078, T2-0083–0087, T4-0007–0013

08. [A] Grating density and spectral resolution / 光栅刻线密度与光谱分辨率

EN: For the same illuminated width and diffraction order, a higher groove density illuminates more grooves N. Resolving power $R=\lambda/\Delta\lambda=mN$ increases, and the minimum resolvable wavelength difference Δλ decreases. Thus 1200 lines/mm resolves more closely spaced lines than 300 lines/mm under these assumptions.

中文： 在受照宽度与衍射级次相同的条件下，刻线密度更高使受照刻线数N增加。分辨本领$R=\lambda/\Delta\lambda=mN$增大，最小可分辨波长差Δλ减小。因此在这些假设下，1200 lines/mm比300 lines/mm能分开更接近的谱线。

EN: Watch the wording: The lecturer explicitly revisits this comparison as a Quiz item. Higher resolving power and smaller Δλ both describe better resolution. The 300/1200 example is anchored to the Week 5 oral review, not an invented Week 2 slide page.

中文： 读题易错点：老师明确说会在Quiz再问该比较。R更大与Δλ更小都表示分辨率更好。300/1200示例依据Week5口头回顾，不虚标为Week2某页原图。

Source / 来源：W2 p35 (spectrometer structure background; no 300/1200 comparison on supplied slides); Sample quiz p1 Q6; Sample quiz p2 Q7; W2 p82–88 | T5-0061–0063, T7-0054–0055

09. [A] Labelled versus label-free optical sensing / 有标记与无标记光学传感

EN: Labelled sensing attaches a reporter such as a fluorescent dye. Label-free sensing detects an intrinsic change such as absorption, refractive index, or resonance shift without adding that reporter. In the sample classification, fluorescence intensity is treated as a labelled method.

中文： 有标记传感添加荧光染料等报告物。无标记传感不添加这种报告物，而测量吸收、折射率或共振位移等内在变化。样题分类把fluorescence intensity视为有标记方法。

EN: Watch the wording: Label-free does not mean no illumination, no biological recognition, or no sensor surface. Autofluorescence exists, so the course classification should not be turned into an absolute claim about every fluorescence measurement.

中文： 读题易错点：Label-free不等于不照光、不需生物识别或不需传感界面。自发荧光确实存在，因此不能把课程分类扩大为所有荧光测量的绝对规律。

Source / 来源：W3 PDF p13–18 | T3-0016–0020

10. [B] Optical fibers and total internal reflection / 光纤与全内反射

EN: A conventional guiding fiber has a higher-index core and lower-index cladding. Total internal reflection requires light travelling from higher to lower refractive index and an incidence angle above the critical angle. Point sensing measures locally; multiplexed sensing uses several sites; distributed sensing reads along the fiber.

中文： 常规导光光纤的纤芯折射率高于包层。全内反射要求光从高折射率介质到低折射率介质，且入射角大于临界角。Point sensing在局部测量；multiplexed sensing有多个位置；distributed sensing沿光纤分布测量。

EN: Watch the wording: An evanescent field outside the guiding region can interact with the surrounding sample. Total internal reflection does not mean the electromagnetic field is exactly zero outside the core.

中文： 读题易错点：导光区域外的evanescent field可与周围样品相互作用。全内反射并不表示纤芯外电磁场严格为零。

Source / 来源：W3 PDF p19–24; W3 PDF p23–28; W3 PDF p31–38 | T3-0030–0045, T3-0052–0057

11. [A] Resonators, SPR and LSPR / 共振器、表面等离激元与局域表面等离激元

EN: Optical resonators enhance repeated light–sample interaction. SPR is a light-coupled collective electron oscillation at a metal interface; LSPR is localized around metal nanoparticles or nanostructures. Surface binding can shift a resonance because the surrounding refractive index changes.

中文： 光学共振器增强光与样品反复相互作用。SPR是金属界面上与光耦合的集体电子振荡；LSPR局域于金属纳米颗粒或纳米结构。表面结合改变周围折射率，因此可能引起共振位移。

EN: Watch the wording: The course uses a broad resonance-sensing classification. Do not equate every SPR structure with a closed Fabry–Perot cavity. For this Quiz, identify mechanism and shift direction before studying long resonance derivations.

中文： 读题易错点：课程采用广义“共振传感”分类，但不能把所有SPR结构等同于闭合Fabry–Perot腔。这次Quiz优先辨识机制与位移，再考虑长共振公式。

Source / 来源：W3 PDF p39–45; W3 PDF p51–60; W3 PDF p64–75 | T3-0062–0072, T3-0084–0093, T3-0099–0105

12. [A] Antigen rapid test (ART): gold nanoparticle LSPR / 抗原快速检测：金纳米颗粒LSPR

EN: The lecturer explicitly says the ART principle is a recurring Quiz question. In the taught colorimetric rapid-test example, gold nanoparticles produce the visible optical response through LSPR; antibody–antigen recognition controls their localization at the test line. The control line checks assay operation; a missing control line makes the test invalid.

中文： 老师明确说ART原理是反复出现的Quiz问题。课程的比色快检示例中，金纳米颗粒通过LSPR产生可见光学响应，抗体—抗原识别控制其在检测线处的聚集或定位。 Control line控制线用于检查检测运行；缺少控制线时结果无效。

EN: Watch the wording: Separate recognition from readout: antibody binding supplies specificity, while the gold nanoparticles supply the optical signal. Do not answer PCR, electrical membrane potential, or ordinary fluorescence excitation for this taught example.

中文： 读题易错点：区分识别与读出：抗体结合赋予特异性，金纳米颗粒提供光学信号。该课程示例不能答成PCR、细胞膜电位或普通荧光激发。

Source / 来源：W3 PDF p76; W7 p52–55 (context; classroom question diagram); W3 p76–79; W1 PDF p67 | T1-0062–0064, T3-0106, T7-0047–0050

13. [A] Numerical aperture and microscope resolution / 数值孔径与显微分辨率

EN: Numerical aperture is NA=n sinθ. At a fixed wavelength, the diffraction-limited lateral separation scales as d≈0.61λ/NA. Larger NA or shorter wavelength gives a smaller resolvable separation. Magnification alone does not guarantee improved resolving ability.

中文： 数值孔径NA=n sinθ。波长固定时，衍射极限横向最小间距约d≈0.61λ/NA。NA更大或波长更短，使可分辨间距更小。只有放大倍率增加，并不保证分辨能力提高。

EN: Watch the wording: The objective-lens comparison is explicitly named again for the Quiz in Week 5. If NA doubles with the same wavelength and criterion, d halves. “Higher resolution” means a smaller minimum distinguishable distance.

中文： 读题易错点：Week5明确说Quiz会再问物镜比较。波长与判据相同、NA加倍时，d减半。“Higher resolution”表示最小可分辨距离更小。

Source / 来源：W4 p27–30; W4 p29–34; W4 p29–30 | T4-0030–0038, T5-0059–0061

14. [A] Confocal and two-photon microscopy / 共聚焦与双光子显微镜

EN: A confocal pinhole suppresses out-of-focus light and enables optical sectioning. Two-photon excitation is nonlinear and largely confined to the focus; two longer-wavelength photons together supply the excitation energy. These mechanisms improve contrast and sectioning in different ways.

中文： 共聚焦pinhole抑制离焦光，实现光学切片。双光子激发具有非线性，主要限制在焦点；两个长波长光子共同提供激发能量。两种机制以不同方式改善对比度与切片能力。

EN: Watch the wording: Do not call a pinhole the cause of two-photon excitation. The usual two-photon wavelength advantage concerns penetration and localized excitation; it does not erase the dependence of diffraction resolution on wavelength and NA.

中文： 读题易错点：不要把pinhole说成双光子激发的原因。双光子的常见优势是穿透与局域激发，并不取消衍射分辨率对波长和NA的依赖。

Source / 来源：W4 p47–61; Sample quiz p1 Q1–3; W2 p73–78; W3 p39–50; W4 p54–61 | T4-0062–0069, T7-0051–0053

15. [A] OCT: low-coherence interference and reflectivity / OCT：低相干干涉与反射率

EN: Optical coherence tomography combines light returning from a sample with a reference beam. Low-coherence interference localizes returning signals in depth. Layer brightness represents back-reflection/back-scattering under the course description; it is not simply a temperature or fluorescence map.

中文： 光学相干断层成像把样品返回光与参考光合并。低相干干涉把返回信号定位到不同深度。按课程描述，层状亮度反映后向反射或散射，而不是简单的温度图或荧光图。

EN: Watch the wording: Identify the reference arm, sample arm and interference signal. The long Michelson intensity derivation is lower priority because of the global short-calculation guidance; there is no separate explicit oral exclusion of all OCT equations.

中文： 读题易错点：要识别reference arm、sample arm和interference signal。依据本次短计算的总要求，Michelson长强度推导降低优先级；没有单独找到“所有OCT公式均不考”的明确声明。

Source / 来源：W4 p71–79; W4 p85–90; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0088–0098, T4-0105–0108, T7-0055–0057

16. [A] CCD, CMOS and hyperspectral imaging / CCD、CMOS与高光谱成像

EN: A CCD accumulates photocharge and transfers it through a charge-transfer readout structure. CMOS commonly uses pixel-associated circuitry for readout or amplification. Hyperspectral imaging records a spectrum at each spatial location, producing a spatial–spectral data cube.

中文： CCD累积光生电荷，并通过电荷转移结构读出；CMOS通常通过像素相关电路读出或放大。Hyperspectral imaging在每个空间位置记录光谱，形成空间—光谱数据立方。

EN: Watch the wording: A line-scan instrument records one spatial line plus its spectra at a time; scanning the other spatial axis builds the cube. For sample Q9, B remains a reasoned reference answer; the oral review does not give an unambiguous final letter.

中文： 读题易错点：Line-scan一次记录一条空间线及其光谱，再沿另一空间轴扫描形成数据立方。样题Q9仍把B作为推导参考答案，但口头回顾没有给出明确的最终字母。

Source / 来源：W4 p35–38; W4 p91–104; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T4-0048–0055, T4-0109–0113, T7-0055–0057

17. [B] Ultrasound versus photoacoustic imaging / 超声与光声成像

EN: Conventional ultrasound uses acoustic transmission and returning echoes. Photoacoustic imaging starts with optical absorption, followed by heating, thermoelastic expansion and acoustic-wave generation; an ultrasound transducer detects the sound. Optical absorption supplies contrast, while acoustic detection supports depth.

中文： 常规超声发射声波并接收回波。光声成像从光吸收开始，随后升温、热弹性膨胀并产生声波，由超声换能器检测。光学吸收提供对比，声学检测支持较深成像。

EN: Watch the wording: Echo depth uses a round-trip time: z=vt/2. For a photoacoustic one-way propagation time, z=vt. The problem must state which timing it measures. Detailed ultrasound scan and resolution examples are lower priority under the local lecture exclusions.

中文： 读题易错点：回波深度按往返时间z=vt/2；若给光声单程传播时间，则z=vt。必须先看题目定义的时间。根据课堂局部排除，复杂超声扫描与分辨率例题降低优先级。

Source / 来源：W5 p5–14; W5 p15–18; W5 p27–28; W5 p29; W5 p34–43 | T5-0004–0017, T5-0037–0040, T5-0044–0051

18. [B] Electrochemical sensing modes / 电化学传感模式

EN: Potentiometric sensing reads potential, ideally with negligible measurement current. Amperometric sensing measures current at a controlled potential. Conductometric sensing reads conductance; impedimetric sensing reads the response to an AC perturbation. Identify the measured variable before doing arithmetic.

中文： Potentiometric测电位，理想情况下测量电流极小；amperometric在控制电位下测电流；conductometric测电导；impedimetric测交流扰动的响应。先识别被测量，再计算。

EN: Watch the wording: Working, reference and counter electrodes have different roles. A reference electrode provides a stable potential reference. In a three-electrode arrangement, the counter electrode carries the balancing current while the reference ideally carries negligible current.

中文： 读题易错点：Working、reference和counter electrode作用不同。参比电极提供稳定电位基准。在三电极结构中，对电极承担配平电流，参比电极理想情况下电流极小。

Source / 来源：W6 p13–18; W6 p19–21; W6 p37–40 | T6-0011–0022, T6-0038–0040

19. [B] Glucose biosensor: recognition and electrical readout / 葡萄糖传感器：识别与电读出

EN: In the taught glucose-oxidase example, an enzymatic reaction links glucose to an electroactive product. An amperometric electrode converts the product-related reaction into current. A paired background channel can help subtract non-target electroactive interference under matched conditions.

中文： 课程的glucose oxidase示例中，酶反应把葡萄糖与电活性产物联系起来。安培型电极把产物相关反应转为电流。条件匹配时，配对背景通道可帮助扣除非目标电活性干扰。

EN: Watch the wording: The enzyme recognition step and electrode transduction step are different. The lactate example is called exam-style without a clear Quiz/final distinction. This edition lowers the multi-step mass-to-moles-to-electrons calculation because of the Quiz’ s short-calculation guidance; keep the simple reaction–charge–current chain as supporting knowledge.

中文： 读题易错点：酶识别与电极换能是不同步骤。乳酸例题被称为考试类型，但没有明确区分Quiz与期末。本版依据Quiz短计算指导降低质量→摩尔→电子多步计量的优先级，保留反应、电荷与电流的简单关系作为辅助知识。

Source / 来源：W6 p55, p57–59; W6 p58–59; W6 p60 | T6-0060–0073

20. [A] ECG, EMG and EEG: match the organ and signal / ECG、EMG与EEG：组织来源对应

EN: ECG records cardiac electrical activity; EMG records muscle electrical activity; EEG records scalp potential differences associated with brain activity. These are biopotentials measured relative to a reference. A smartwatch can estimate sleep from motion and heart-related signals without directly recording scalp EEG.

中文： ECG记录心脏电活动，EMG记录肌肉电活动，EEG记录与脑活动有关的头皮电位差。它们都是相对于参照测得的生物电位。智能手表可以用运动与心脏相关信号估计睡眠，而不直接记录头皮EEG。

EN: Watch the wording: Do not infer that every wearable measures EEG. The sample smartwatch combination is accelerometer plus heart-rate/heart-rate-variability information. EMG numerical ranges labelled “for reference” do not remove basic EMG recognition from the review.

中文： 读题易错点：不能认为所有可穿戴设备都测EEG。样题手表对应accelerometer加heart rate/heart rate variability。EMG某些数值范围被标为“供参考”，并不等于基础EMG识别全部不用复习。

Source / 来源：W7 p5–8; Sample quiz p1 Q4; W1 p16, p48, p69–75 | T7-0002–0006, T7-0053

21. [B] Membrane potential and action-potential phases / 膜电位与动作电位阶段

EN: Use V_m=Vinside−Voutside. A resting membrane is usually negative inside. Depolarization means becoming less negative and may include a positive overshoot; repolarization returns toward rest; hyperpolarization becomes more negative than rest. The taught sequence emphasizes Na^+ influx and subsequent K^+ efflux.

中文： 使用V_m=Vinside−Voutside。静息时细胞内通常为负。Depolarization表示变得不那么负，并可能出现正向超射；repolarization返回静息；hyperpolarization比静息更负。课程重点顺序为Na^+内流，随后K^+外流。

EN: Watch the wording: “The inside is always negative” is false during an action potential. Keep mV separate from V. Long multi-ion membrane equations and derivations are secondary to the phase names, ion directions and sign convention.

中文： 读题易错点：“细胞内始终为负”在动作电位中是错误的。mV与V不能混。长的多离子膜方程及推导，优先级低于阶段名称、离子流向和符号约定。

Source / 来源：W7 p22–27; W7 p19–21; W7 p32–39 | T7-0011–0012, T7-0014–0018, T7-0023–0025, T7-0028–0030

22. [A] ECG morphology and reading frequency from a trace / ECG波形与从时轴读频率

EN: The P wave relates to atrial depolarization, the QRS complex to ventricular depolarization, and the T wave to ventricular repolarization. For a reasonably periodic trace, frequency f=N/Δt or f=1/T. Always read the horizontal-axis unit first.

中文： P wave对应心房去极化，QRS complex对应心室去极化，T wave对应心室复极化。近似周期波形的频率f=N/Δt或f=1/T。必须先读横轴单位。

EN: Watch the wording: Sample Q5 is a rough estimate from a low-resolution, irregular EEG trace, around 45 Hz with C (40–48 Hz) only a provisional reference, not a confirmed answer. Treat it as graph-reading practice rather than an exact numerical standard.

中文： 读题易错点：样题Q5来自低分辨率且不规则的EEG波形，仅能估读约45 Hz，C（40–48 Hz）仅作暂定参考，不是已确认答案。应把它作为读图训练，而非精确数值标准。

Source / 来源：W7 p62–69; W7 p79–87; Sample quiz p1 Q5 | T7-0068–0073, T7-0084–0087, T7-0091–0095

23. [A] Patch clamp versus multielectrode array (MEA) / 膜片钳与多电极阵列

EN: Patch clamp forms a tight seal between a micropipette and a membrane patch and can record controlled membrane currents or potentials depending on the mode. A MEA records extracellular signals at many electrode sites in parallel. The lecturer explicitly names these two methods for Quiz/exam recognition.

中文： Patch clamp用微吸管与膜片形成紧密封接，根据模式记录受控的膜电流或电位。MEA通过多个电极位点并行记录细胞外信号。老师明确点名这两种方法可作为Quiz或考试的辨识内容。

EN: Watch the wording: Recognize the pipette–membrane seal versus the patterned electrode array. Do not replace a standard electrode array with the excluded detailed conformal-electrode material/design requirements.

中文： 读题易错点：图中要分清微吸管—细胞膜封接与阵列化电极。基础MEA辨识不能被替换成已排除的贴合电极材料或技术设计细节。

Source / 来源：W8 p9–13; W8 p9–10; W8 p11–13 | T8-0005–0016, T8-0018

24. [B] Differential amplification and common-mode noise / 差分放大与共模噪声

EN: A differential amplifier amplifies the difference between two inputs: $V_{out}=G(V_1-V_2)$. A signal appearing similarly at both inputs is common-mode and is ideally rejected. High input impedance reduces loading of a weak biological signal.

中文： 差分放大器放大两输入之差：$V_{out}=G(V_1-V_2)$。两端相似的信号是common-mode，理想情况下被抑制。较高输入阻抗减少对微弱生物信号的负载影响。

EN: Watch the wording: Subtract before multiplying, and convert mV to V correctly. The reference and recording locations matter. Gain alone does not guarantee removal of noise that appears differently at the two inputs.

中文： 读题易错点：先相减再乘增益，并正确换算mV与V。记录位置和参考位置很重要。若噪声在两输入端不同，仅有增益并不保证它被消除。

Source / 来源：W7 p52–54; W8 p19–20; W8 p20; W8 p22 | T7-0043–0046, T8-0021–0026

25. [B] High-pass, low-pass, band-pass and notch filters / 高通、低通、带通与陷波滤波器

EN: A high-pass filter attenuates low-frequency components; a low-pass filter attenuates high-frequency components. A band-pass filter preserves a frequency interval. A band-reject or notch filter suppresses an interval or narrow interference band, such as mains interference when appropriately selected.

中文： High-pass抑制低频，low-pass抑制高频；band-pass保留一个频率区间；band-reject或notch抑制一个区间或窄干扰带，例如适当选择时可抑制工频干扰。

EN: Watch the wording: Keep filter-role recognition in the main review. The particular RC cutoff example was excluded from this year’ s exam in its local context; this does not justify excluding every filter concept from the Quiz.

中文： 读题易错点：主复习保留滤波功能辨识。某个RC截止频率例题在局部语境中被说成今年考试不考，这不能推导为所有滤波概念都不在Quiz范围。

Source / 来源：W8 p25–26; W8 p27–28; W8 p29 | T8-0026–0029, T8-0062–0063

B: breadth check, Weeks 1–8 / B：逐周概念补全

EN: These 32 short entries replace the previous six broad summaries. Read every entry once: lower priority for a derivation does not remove its basic concept. They extend recognition coverage beyond the 25 mock questions; use the A/B cards for deeper revision.

中文： 以下32条替代之前六段粗略补充。每条至少看一遍：降低某推导的优先级，不代表删掉对应基础概念。它们补足25道模拟题之外的辨识范围；深入复习仍以A/B重点卡为主。

B01 • W1 • Biorecognition and sensor families / 生物识别与传感器类别

EN: Antibodies bind antigens; enzymes catalyse reactions; complementary nucleic acids hybridize. Recognition supplies selectivity, while the transducer supplies readout. Optical sensors read light, electrochemical sensors read electrical changes, and mechanical sensors can read mass-induced displacement or frequency shifts. AI interprets measured data; it does not replace the physical sensing step.

中文： Antibody抗体结合抗原；enzyme酶催化反应；互补核酸通过hybridization杂交识别。识别环节提供选择性，换能器提供读出。光学传感测光，电化学传感测电变化，机械传感可测质量引起的位移或频移。AI分析测得的数据，不能替代实际感测。

Source / 来源：W1 PDF p1–3; W1 PDF p21–31; W1 PDF p39–42; W1 PDF p43–44; W1 PDF p45–49; W1 PDF p50–57; W1 PDF p68–79 | T1-0001–0004, T1-0022–0027, T1-0033–0052, T1-0065–0072

B02 • W1 • Performance beyond sensitivity / 灵敏度之外的性能指标

EN: Linearity describes how closely output follows a straight calibration line. Response time describes how quickly output approaches its final value (95% in these slides). Reproducibility describes agreement across repeated measurements. Selectivity concerns interferents; sensitivity is calibration slope; LOD combines slope with noise. None of these terms alone means overall accuracy.

中文： Linearity线性度描述输出接近直线校准关系的程度；response time响应时间描述接近最终值的速度（课件取95%）；reproducibility再现性描述重复测量的一致程度。Selectivity选择性针对干扰物；sensitivity灵敏度是校准斜率；LOD兼顾斜率与噪声。任何一个指标都不单独等于整体准确性。

Source / 来源：W1 PDF p58–61; W1 PDF p60–62; W1 PDF p63–66 | T1-0053–0061

B03 • W1 • POCT, microfluidics and the sensing workflow / 现场检测、微流控与检测流程

EN: Point-of-care testing (POCT) brings testing near the patient; centralized testing sends samples to a laboratory. Sampling → recognition → transduction → analysis is the basic workflow. Microfluidic biochips manipulate small fluid volumes and support integration and automation; wearable devices support repeated or continuous monitoring. Portability alone does not guarantee better analytical performance.

中文： Point-of-care testing（POCT，即时/现场检测）把检测带到患者附近；centralized testing把样品送到中心实验室。基本流程为采样→识别→换能→分析。Microfluidic biochip微流控芯片操控少量液体，便于集成与自动化；wearable可穿戴设备支持重复或连续监测。便携本身不保证分析性能更好。

Source / 来源：W1 PDF p32–36; W1 PDF p63–66; W6 p3–12; Sample quiz p2 Q8–9; W4 p35–38, p73–86; W3 p25–32 | T1-0028–0031, T1-0059–0061, T6-0001–0010, T7-0055–0057

B04 • W2 • Light models, sources and coherence / 光的模型、光源与相干性

EN: Rays describe paths, waves explain interference/diffraction, and photons describe energy exchange. Lasers commonly offer narrower spectra and greater coherence than LEDs or lamps; CW means continuous-wave, while pulsed means time-separated emission. Coherence is a stable phase relationship. Source power and noise affect usable signal; more power is not always better because saturation or photodamage can occur.

中文： Ray光线模型描述路径，wave波模型解释干涉/衍射，photon光子模型描述能量交换。激光通常比LED或灯具具有更窄光谱和更高相干性；CW为连续波，pulsed为脉冲发射。Coherence相干性指稳定相位关系。光源功率与噪声影响有效信号；功率过大可能饱和或光损伤，并非越大越好。

Source / 来源：W2 PDF p13–22; W2 PDF p21–28 | T2-0010–0027

B05 • W2 • Absorption, scattering and penetration / 吸收、散射与穿透

EN: Absorption transfers optical energy to matter; scattering redirects light. Ballistic attenuation can include both: μt=μa+μs. Rayleigh and Mie scattering are elastic; their regimes depend on particle size relative to wavelength. Raman scattering is inelastic. Tissue penetration depends on both absorption and scattering; the near-infrared window is useful, but not loss-free.

中文： Absorption吸收把光能传给物质；scattering散射改变光的传播方向。直达光衰减可同时包含两者：μt=μa+μs。Rayleigh瑞利与Mie米氏散射都是弹性散射，其适用情形取决于颗粒相对波长的尺寸；Raman拉曼为非弹性。组织穿透取决于吸收与散射；近红外窗口有利于穿透，但并非没有损耗。

Source / 来源：W2 PDF p29–33; W2 PDF p44–46; W2 PDF p55–62; W2 PDF p61–64 | T2-0028–0036, T2-0052–0055, T2-0070–0082

B06 • W2 • Pulse oximetry / 脉搏血氧测量

EN: Two wavelengths exploit different absorption by oxyhemoglobin and deoxyhemoglobin; the pulsatile component helps isolate the arterial contribution. Know oxygenated versus deoxygenated hemoglobin and differential absorption. This estimates oxygen saturation, not simply the total amount of blood. The long saturation equation is lower priority under the lecturer’ s instruction.

中文： 两个波长利用oxyhemoglobin氧合血红蛋白与deoxyhemoglobin脱氧血红蛋白的吸收差异；pulsatile脉动分量帮助分离动脉贡献。要认得氧合/脱氧及差异吸收。估计量是oxygen saturation血氧饱和度，不是简单的总血量。按老师说明，长的饱和度方程降低优先级。

Source / 来源：W2 PDF p47–54 | T2-0056–0069

B07 • W2 • Luminescence and state transitions / 发光类别与能级跃迁

EN: Fluorescence usually emits from S1 to S0; phosphorescence involves a triplet state and is generally slower. Internal conversion is nonradiative relaxation between states of the same spin multiplicity; intersystem crossing changes multiplicity. Chemiluminescence uses chemical excitation, and bioluminescence is biological chemiluminescence. An emission peak need not have the same wavelength as absorption.

中文： Fluorescence荧光通常由S1到S0发射；phosphorescence磷光涉及三重态，通常更慢。Internal conversion内转换是在相同自旋多重度能态间的非辐射弛豫；intersystem crossing系间窜越改变多重度。Chemiluminescence化学发光由化学反应激发，bioluminescence生物发光属于生物化学发光。发射峰不必与吸收峰波长相同。

Source / 来源：W2 PDF p69–76; W2 PDF p77–79; W2 PDF p82–88 | T2-0088–0100, T2-0105–0107

B08 • W2 • Raman shift and spectral axes / 拉曼位移与光谱坐标

EN: Wavelength (nm), frequency (Hz) and wavenumber (cm^{-1}) are different axes. For Stokes scattering, $\Delta\tilde{\nu}=10^{7}(1/\lambda exc-1/\lambda S)$ cm^{-1} when both wavelengths are in nm. A 1000 nm excitation and 1080 nm Stokes line give about 741 cm^{-1}. Anti-Stokes photons have higher energy than excitation; Stokes photons have lower energy.

中文： Wavelength波长（nm）、frequency频率（Hz）、wavenumber波数（cm^{-1}）是不同坐标。两波长均用nm时，斯托克斯位移$\Delta\tilde{\nu}=10^{7}(1/\lambda exc-1/\lambda S)$，单位$cm^{-1}$。1000 nm激发、1080 nm斯托克斯线对应约741 cm^{-1}。Anti-Stokes反斯托克斯光子能量高于激发，Stokes斯托克斯光子能量低于激发。

Source / 来源：W2 PDF p34–35; W2 PDF p55–62; W2 PDF p65–68 | T2-0037–0040, T2-0070–0078, T2-0083–0087

B09 • W3 • Quantum dots and flow cytometry / 量子点与流式细胞术

EN: Quantum dots are semiconductor fluorescent labels with size-dependent emission; larger dots commonly emit at longer wavelengths within the same material family. Flow cytometry measures cells/particles passing a detection region: forward scatter (FSC) mainly indicates size, side scatter (SSC) internal complexity/granularity, and fluorescence label-associated molecular information. A scatter plot separates populations; it is not a tissue image.

中文： Quantum dot量子点是半导体荧光标记，发射随尺寸变化；同材料系列中，较大颗粒通常发射更长波长。Flow cytometry流式细胞术测量流经检测区的细胞/颗粒：FSC前向散射主要反映大小，SSC侧向散射反映内部复杂度/颗粒度，荧光反映标记所对应的分子信息。散点图用于区分群体，不是组织图像。

Source / 来源：W2 PDF p89–95; W3 PDF p5–9; W3 PDF p10–12 | T2-0108–0111, T3-0003–0015

B10 • W3 • Fibre modes and cavity recognition / 光纤模式与共振腔辨识

EN: A mode is an allowed field pattern; single-mode and multimode fibres differ in supported patterns. Evanescent fields decay outside the guiding region and enable surface sensing. Fabry–Pérot cavities use facing reflectors; whispering-gallery modes circulate near a boundary; photonic-crystal cavities confine light using a structured refractive-index arrangement. Multiplexing uses distinct sensing sites; distributed sensing samples along the fibre.

中文： Mode模式是允许存在的场分布；single-mode单模与multimode多模光纤支持的模式数不同。Evanescent field倏逝场在导光区域外衰减，可用于表面传感。Fabry–Pérot腔使用相对反射面；whispering-gallery mode回音壁模式沿边界循环；光子晶体腔利用结构化折射率分布约束光。Multiplexed在多个离散位点检测；distributed沿光纤分布检测。

Source / 来源：W3 PDF p19–24; W3 PDF p23–28; W3 PDF p30; W3 PDF p31–38; W3 PDF p31–39; W3 PDF p39–45; W3 PDF p48–50 | T3-0030–0045, T3-0051–0072, T3-0079–0083

B11 • W3 • Sensitivity, linewidth, Q and FOM / 灵敏度、线宽、Q值与品质指标

EN: Bulk refractive-index sensitivity $S\lambda=\Delta\lambda/\Delta n$ has units nm/RIU. It describes index response, not biochemical selectivity. Resonance linewidth is often FWHM (full width at half maximum); $Q=f0/\delta f$, equivalently $\lambda0/\delta\lambda$ for a narrow resonance. $FOM\lambda=S\lambda/\delta\lambda$ compares sensitivity with linewidth. Keep the same spectral coordinate and units; narrower lines alone do not determine LOD.

中文： 体折射率灵敏度$S\lambda=\Delta\lambda/\Delta n$，单位nm/RIU，描述折射率响应而非生化选择性。共振线宽常用FWHM半高全宽；$Q=f0/\delta f$，窄共振下等价于$\lambda0/\delta\lambda$。$FOM\lambda=S\lambda/\delta\lambda$同时考虑灵敏度与线宽。分子分母必须使用相同光谱坐标与匹配单位；线更窄也不能单独确定LOD。

Source / 来源：W3 PDF p15–18; W3 PDF p25–29; W3 PDF p46–50; W3 PDF p77–80; W3 PDF p81–87 | T3-0021–0029, T3-0046–0050, T3-0073–0078, T3-0107–0114

B12 • W3 • LSPR tuning and SERS / 局域等离激元调谐与表面增强拉曼

EN: Nanoparticle material, size, shape and surrounding refractive index affect LSPR. SERS means surface-enhanced Raman scattering: metal nanostructures create strong local fields, especially at hot spots, and greatly enhance nearby Raman signals. Recognize a Raman molecular fingerprint versus an LSPR resonance shift. SERS was continued in Week 4; its deferral in Week 3 did not remove it from scope.

中文： 纳米颗粒材料、尺寸、形状和周围折射率影响LSPR。SERS为surface-enhanced Raman scattering表面增强拉曼散射：金属纳米结构产生强局域场，尤其在hot spot热点处，显著增强附近拉曼信号。区分拉曼分子指纹与LSPR共振位移。SERS在Week4续讲，Week3推迟讲解不表示不考。

Source / 来源：W3 PDF p64–75; W3 PDF p88–95; W3 p60–94; W3 p88–94 | T3-0099–0105, T3-0115–0116, T4-0001–0014

B13 • W4 • Imaging modality identification / 成像技术辨识

EN: X-ray imaging measures transmission attenuation; CT reconstructs many X-ray projections. MRI uses magnetic fields and radiofrequency excitation; PET means positron emission tomography and uses a radioactive tracer. Ultrasound uses sound; optical imaging uses light. X-rays/CT and PET involve ionizing radiation. Compare contrast mechanism, penetration and resolution rather than assuming one method is best for every task.

中文： X-ray成像测透射衰减；CT由多个X射线投影重建。MRI磁共振使用磁场及射频激发；PET为positron emission tomography正电子发射断层，使用放射性示踪剂。Ultrasound超声用声，optical imaging光学成像用光。X射线/CT和PET涉及电离辐射。比较对比机制、穿透与分辨率，不能认定某技术对所有任务都最好。

Source / 来源：W4 p3–12; W5 p53–58; W5 p64–69 | T4-0015–0021, T5-0073–0080, T5-0088–0095

B14 • W4 • Optical biopsy, endoscopy and virtual staining / 光学活检、内窥镜与虚拟染色

EN: Optical biopsy seeks tissue information through optical imaging, reducing tissue removal in the taught examples. Endoscopy brings illumination and detection to internal sites, often with fibre delivery. Virtual staining computationally makes an image resemble a stained tissue image; it does not physically add a dye. Recognize the workflow without assuming every optical method replaces histopathology.

中文： Optical biopsy光学活检通过光学成像获取组织信息，在课程示例中减少取组织操作。Endoscopy内窥镜把照明与检测带到体内，常用光纤传输。Virtual staining虚拟染色用计算产生类似染色组织的图像，并未实际添加染料。要认得流程，不应推断所有光学方法都能替代组织病理。

Source / 来源：W4 p13–20; W4 p65–70; W4 p106–112 | T4-0022–0025, T4-0085–0087, T4-0115–0120

B15 • W4 • NA, Airy disk and detector sampling / NA、艾里斑与探测器采样

EN: In NA=n sinθ, θ is the collection cone half-angle; immersion increases n. Rayleigh separation ≈0.61λ/NA is different from Airy-disk diameter ≈1.22λ/NA. Pixels must sample optical features sufficiently after magnification. More pixels cannot restore detail absent from the optical image. The thin-lens relation 1/f=1/u+1/v uses the stated lens sign convention.

中文： NA=n sinθ中，θ为收集锥半角；浸液可提高n。Rayleigh最小分辨间距约0.61λ/NA，与Airy-disk艾里斑直径约1.22λ/NA不同。放大后的光学细节需要充分像素采样；增加像素无法恢复光学成像中本来缺失的细节。薄透镜式1/f=1/u+1/v需配合题目规定的符号约定。

Source / 来源：W4 p21–26; W4 p27–30; W4 p29–34; W4 p31–34 | T4-0026–0038, T4-0045–0047

B16 • W4 • Photodiode and photomultiplier tube / 光电二极管与光电倍增管

EN: A photodiode converts absorbed photons into electron–hole pairs and photocurrent. A PMT emits photoelectrons from a photocathode, then multiplies electrons at successive dynodes. This internal gain helps detect weak light; it does not remove all noise. A point detector usually requires scanning for an image, whereas CCD/CMOS arrays provide many spatial detector elements.

中文： Photodiode光电二极管将吸收光子转换为电子—空穴对及光电流。PMT光电倍增管由photocathode光阴极发出光电子，再在dynode倍增极逐级倍增电子。内部增益帮助检测弱光，不等于消除所有噪声。点探测器通常需扫描才能成像；CCD/CMOS阵列提供许多空间探测单元。

Source / 来源：W4 p35–38 | T4-0048–0055

B17 • W4 • Sectioning, OCT resolution and hyperspectral data / 切片、OCT分辨率与高光谱数据

EN: Widefield collects out-of-focus background; confocal rejects it; two-photon excitation is localized (fluorescence $\propto I^2$ before saturation). OCT axial resolution depends on coherence length: broader bandwidth improves it at fixed central wavelength. Lateral resolution depends on focusing/NA; higher NA reduces depth of focus, the usable focused depth range, not layer separation. Hyperspectral imaging records many narrow bands at each position, not just RGB.

中文： Widefield宽场会收集离焦背景，共聚焦抑制它，双光子采用局域激发（未饱和时荧光$\propto I^2$）。OCT轴向分辨率取决于相干长度：中心波长固定时，带宽越宽、轴向分辨越好。横向分辨率依赖聚焦/NA；NA越大、焦深越小。Depth of focus焦深是可用聚焦深度范围，不是可分辨层间距。高光谱在每个位置记录许多窄谱带，不只是RGB。

Source / 来源：W4 p39–46; W4 p47–61; W4 p71–79; W4 p80–82; W4 p83; W4 p91–104; W4 p105 | T4-0056–0069, T4-0088–0104, T4-0109–0114

B18 • W5 • Piezoelectric transmission, echoes and attenuation / 压电发射、回波与衰减

EN: Inverse piezoelectric effect: applied voltage produces deformation and sound (transmit). Direct effect: returning pressure produces charge (receive). Acoustic impedance Z=ρc; a larger mismatch produces stronger reflection, so coupling gel reduces the air gap. Higher ultrasound frequency usually improves spatial resolution but increases attenuation and reduces penetration. Echo attenuation uses the outgoing and returning path.

中文： Inverse piezoelectric effect逆压电效应：施加电压产生形变与声波（发射）；direct正压电效应：返回声压产生电荷（接收）。声阻抗Z=ρc；失配越大、反射通常越强，耦合凝胶用于减少空气间隙。超声频率更高通常提高空间分辨率，但增加衰减、降低穿透。回波衰减应计入去程和回程。

Source / 来源：W5 p5–14; W5 p15–18; W5 p19–23; W5 p19–21; W5 p24–26; W5 p27–28; W5 p29 | T5-0004–0032, T5-0037–0040

B19 • W5 • Photoacoustic contrast and X-ray attenuation / 光声对比与X射线衰减

EN: Photoacoustic contrast depends on optical absorption, so excitation wavelength can emphasize different absorbers. An X-ray tube accelerates electrons toward a target to generate X-ray photons; the patient is imaged by photons, not the electron beam. X-ray transmission follows exponential attenuation; successive layers multiply transmissions. Absorption contrast and acoustic reflection contrast are different mechanisms.

中文： 光声对比依赖光吸收，因此激发波长可突出不同吸收物。X射线管把电子加速到靶材以产生X射线光子；用于成像的是光子，不是射向患者的电子束。X射线透射遵循指数衰减；连续多层的透过率相乘。吸收对比与声学反射对比属于不同机制。

Source / 来源：W5 p34–43; W5 p44–51; W5 p53–58; W5 p59–61; W5 p62; W5 p63 | T5-0044–0051, T5-0063–0087

B20 • W5 • Digital images and analysis vocabulary / 数字图像与分析术语

EN: Sampling discretizes position; quantization discretizes intensity (2^b levels for b bits). Preprocessing includes denoising/deblurring; convolution combines neighbouring values with kernel weights. Semantic segmentation labels classes; instance segmentation separates objects. Classification assigns a class using features such as shape/intensity. TP/TN mean true positive/negative; FP/FN mean false positive/negative. Diagnostic sensitivity=TP/(TP+FN), specificity=TN/(TN+FP); these differ from calibration slope.

中文： Sampling采样离散化位置；quantization量化离散化强度（b位对应2^b级）。Preprocessing预处理含去噪/去模糊；convolution卷积用核权重组合邻域值。Semantic segmentation语义分割标类别；instance segmentation实例分割区分对象。Classification分类依据形状/强度等特征给出类别。TP/TN为真阳性/真阴性，FP/FN为假阳性/假阴性。诊断敏感度=TP/(TP+FN)，特异度=TN/(TN+FP)，不同于校准斜率。

Source / 来源：W5 p70–74; W5 p75–83; W5 p84–86 | T5-0096–0106

B21 • W6 • Redox and reference-relative potential / 氧化还原与相对电位

EN: Oxidation loses electrons; reduction gains electrons. Oxidation occurs at the anode and reduction at the cathode; electrode polarity depends on cell type. Electrode potential is measured relative to a reference. Nernst-type equilibrium potentials depend logarithmically on ion activities, not linearly on concentration. When using reduction-potential tables, Ecell=Ecathode−Eanode; do not mix oxidation and reduction signs.

中文： Oxidation氧化失电子；reduction还原得电子。Anode阳极发生氧化、cathode阴极发生还原，但正负极性取决于电池类型。电极电位相对参比测量。Nernst型平衡电位与离子活度的对数有关，并非与浓度直接成正比。使用还原电位表时Ecell=Ecathode−Eanode，不要混用氧化与还原电位符号。

Source / 来源：W6 p13–18; W6 p31–35, p38; W6 p36; W6 p37–40; W6 p41–46 | T6-0011–0018, T6-0034–0043

B22 • W6 • Microfabrication and PDMS / 微加工与PDMS

EN: Photolithography patterns photoresist using exposure and development; etching removes selected material, whereas deposition adds material. PDMS (polydimethylsiloxane) soft lithography casts and cures a polymer on a master, then peels and bonds it to form structures such as microchannels. Know each step’ s function and the integration advantage; detailed recipes and equipment settings are secondary.

中文： Photolithography光刻通过曝光、显影形成光刻胶图案；etching刻蚀去除选定材料，deposition沉积添加材料。PDMS聚二甲基硅氧烷软光刻在母模上浇注、固化聚合物，再剥离、键合，可形成微通道。认识各步骤功能与集成优势即可，详细配方和设备参数次要。

Source / 来源：W6 p22–30 | T6-0023–0033

B23 • W6 • Three glucose-sensor generations / 葡萄糖传感器三代原理

EN: First generation uses the natural oxygen/peroxide pathway; second generation uses an electron-transfer mediator; third generation aims at direct enzyme–electrode electron transfer. In the taught peroxide-oxidation example: $H_2O_2 \rightarrow O_2+2H^+ +2e^-$. A matched electrode without GOx measures background for subtraction. Do not generalize this example into “all amperometric sensors must measure peroxide”.

中文： 第一代使用天然氧气/过氧化氢途径；第二代使用electron-transfer mediator电子传递介体；第三代追求酶—电极直接电子传递。课程的过氧化氢氧化例子为$H_2O_2 \rightarrow O_2+2H^+ +2e^-$。匹配的无GOx电极测背景以便相减。不能推广为“所有安培型传感器都必须测过氧化氢”。

Source / 来源：W6 p55, p57–59; W6 p58–59 | T6-0064–0073

B24 • W6 • Charge, current and reaction stoichiometry / 电荷、电流与反应计量

EN: For constant current, Q=It. Faraday’ s relation is $Q=n_eF=z\ n_aF$, where n_e is moles of electrons, n_a moles of analyte, z electrons per analyte molecule and F≈96485 C/mol. Obtain z from the stated reaction. Mass-to-moles requires molar mass; counting molecules requires Avogadro’ s constant. Keep the short relations; lengthy multi-step examples are lower priority.

中文： 恒定电流时Q=It。法拉第关系$Q=n_eF=z\ n_aF$，其中n_e为电子摩尔数，n_a为分析物摩尔数，z为每分子转移电子数，F≈96485 C/mol。z应由给定反应式获得。质量换摩尔需摩尔质量，分子数换算需阿伏伽德罗常数。保留短关系，长的多步例题降低优先级。

Source / 来源：W6 p50–52; W6 p50–52, p60; W6 p60; W7 p9–13 | T6-0054–0063, T7-0006–0008

B25 • W6 • FET, nanopore and capacitive readout / 场效应管、纳米孔与电容读出

EN: In a field-effect transistor (FET) sensor, binding near the gate changes the channel’ s electrical response. A nanopore reads ionic-current changes/blockades as molecules interact with the pore. A capacitive sensor reads changes in capacitance, which depends on geometry and dielectric properties (ideal plates: C=εA/d). Conductance G=1/R; conductance and material conductivity are related but different quantities.

中文： Field-effect transistor（FET，场效应晶体管）传感器中，栅极附近结合改变沟道电响应。Nanopore纳米孔测分子与孔相互作用引起的离子电流变化/阻塞。Capacitive电容型传感测电容变化，电容依赖几何与介电性质（理想平板C=εA/d）。电导G=1/R；conductance电导与conductivity材料电导率相关，但并非同一物理量。

Source / 来源：W6 p64–74; W6 p75–77; W6 p78–81, p87–90 | T6-0074–0091, T6-0102–0105

B26 • W6 • Impedance, reactance and Bode plots / 阻抗、电抗与Bode图

EN: Impedance includes magnitude and phase in AC response. Ideal R is frequency-independent; |XC|=1/(2πfC) decreases with frequency, while XL=2πfL increases. A Bode plot shows magnitude and phase versus logarithmic frequency. A capacitor’ s imaginary impedance is not its positive magnitude. Long complex-number combinations and full Bode plotting remain secondary for this Quiz.

中文： Impedance阻抗包含交流响应的幅值与相位。理想R不随频率变；|XC|=1/(2πfC)随频率升高减小，XL=2πfL随频率升高增大。Bode图画幅值和相位随对数频率的变化。电容的虚数阻抗不能等同于其正值幅度。复杂阻抗组合与完整Bode作图仍为本次Quiz次要内容。

Source / 来源：W6 p82–85; W6 p84–86; W6 p86; W7 p70–78 | T6-0092–0101, T7-0073–0081

B27 • W7 • Neurons, threshold and all-or-none firing / 神经元、阈值与全或无发放

EN: Dendrites receive input; the axon conducts action potentials. Beyond threshold, a neuronal action potential is approximately all-or-none; stronger stimulation commonly increases firing rate, not single-spike amplitude proportionally. Below threshold does not mean zero voltage: a typical neuronal resting value is about −70 mV, varying by cell. Passive recording measures activity; active stimulation applies an external electrical input.

中文： Dendrite树突接收输入，axon轴突传导动作电位。达到threshold阈值后，神经元动作电位近似all-or-none全或无；更强刺激通常提高firing rate发放频率，而非成比例增大单个尖峰幅度。未达阈值不等于零电压：典型神经元静息约−70 mV，不同细胞可不同。Passive recording被动记录测活动；active stimulation主动刺激施加外部电输入。

Source / 来源：W7 p5–8; W7 p19–21; W7 p32–39 | T7-0002–0006, T7-0011–0012, T7-0023–0025, T7-0028–0030

B28 • W7 • Ionic gradients: Nernst versus GHK / 离子梯度：Nernst与GHK

EN: Membrane voltage arises from ion gradients and selective permeability: typically more K^+ inside and Na^+ outside. Nernst describes the equilibrium potential for one ion; Goldman–Hodgkin–Katz (GHK) includes several ions weighted by permeability. Zero net membrane current does not require each ionic current to be zero. Learn the meaning of the equations without prioritizing their long derivations.

中文： 膜电压来自离子梯度与选择性通透：通常K^+胞内更多、Na^+胞外更多。Nernst能斯特描述单种离子的平衡电位；Goldman–Hodgkin–Katz（GHK）按通透性考虑多种离子。跨膜净电流为零不要求每种离子电流都为零。记住方程含义，长推导不作为优先任务。

Source / 来源：W7 p15–16; W7 p22–27; W7 p25–30 | T7-0008–0010, T7-0014–0021, T7-0030

B29 • W7 • Electrode–electrolyte interface / 电极—电解质界面

EN: Tissue carries ionic current; metal leads carry electronic current. The interface introduces a half-cell potential and frequency-dependent polarization/contact impedance. The course equivalent circuit contains a half-cell source, an interfacial resistance/capacitance branch and solution resistance. Poor or unequal contact can distort recorded signals; a pure series-resistor exercise is only a simplified model.

中文： 组织中为离子电流，金属导线中为电子电流。界面引入half-cell potential半电池电位及随频率变化的极化/接触阻抗。课程等效电路含半电池电源、界面电阻/电容支路及溶液电阻。接触差或两侧不匹配可能使记录失真；纯串联电阻练习只是简化模型。

Source / 来源：W7 p41–42; W7 p52–54; W7 p70–78 | T7-0035–0038, T7-0043–0046, T7-0073–0081

B30 • W7 • ECG leads, heart rate and EMG / 心电导联、心率与肌电

EN: An ECG lead is a potential-difference view; P, QRS and T are features within a waveform, not three different leads. For standard limb leads, II=I+III. Heart rate=60/RR when RR is in seconds; RR=0.8 s gives 75 beats/min. EMG measures muscle electrical activity, with electrode placement and reference affecting the recorded signal.

中文： ECG lead心电导联是一种电位差观察视角；P、QRS、T是同一波形内的特征，不是三个不同导联。标准肢体导联满足II=I+III。RR用秒时，heart rate心率=60/RR；RR=0.8 s对应75次/分。EMG肌电测肌肉电活动，电极位置与参考位置影响记录。

Source / 来源：W7 p47–48; W7 p52–54; W7 p62–69 | T7-0038–0039, T7-0043–0046, T7-0068–0073

B31 • W7 • EEG placement and brain-wave vocabulary / 脑电位置与脑波词汇

EN: Scalp EEG reflects population activity; the 10–20 system standardizes electrode positions. Slow-to-fast order: delta, theta, alpha, beta, gamma. W7 p84 lists delta 1–3, theta 4–7, alpha 8–12, beta 13–25 Hz; exact boundaries vary by convention, and its gamma <25 label is erroneous. Alpha often accompanies relaxed wakefulness, delta slow-wave sleep; these are associations, not diagnoses.

中文： 头皮EEG反映群体活动，10–20 system规范电极位置。由慢到快为delta δ、theta θ、alpha α、beta β、gamma γ。W7 p84列出δ 1–3、θ 4–7、α 8–12、β 13–25 Hz；具体边界随约定变化，其中γ<25为错误标注。Alpha常伴随放松清醒，delta常见于慢波睡眠；这是相关表现，不是诊断规则。

Source / 来源：W7 p79–87; Sample quiz p1 Q5 | T7-0084–0087, T7-0091–0095

B32 • W8 • Amplifier chain, isolation and polarity / 放大链、隔离与极性

EN: A preamplifier handles weak input; high-pass filtering reduces slow drift and low-pass filtering reduces fast interference. Isolation supports patient protection. High input impedance reduces loading; low output impedance helps drive a load. Neither specifies voltage gain. Inverting amplification reverses polarity; non-inverting preserves it. Recognize the ideal gains −Rf/Rin and 1+Rf/Rin as supporting relations, not an explicit memorization promise.

中文： Preamplifier前置放大器处理弱输入；高通减小慢漂移，低通减小快速干扰。Isolation隔离用于患者保护。高输入阻抗减少加载，低输出阻抗便于驱动负载，二者都不直接决定电压增益。Inverting反相放大翻转极性；non-inverting同相放大保持极性。理想增益−Rf/Rin和1+Rf/Rin可作辅助辨识，并非老师明确保证必背。

Source / 来源：W8 p13; W8 p19–20; W8 p20; W8 p21; W8 p19–29 context; W8 p22; W8 p23–24; W8 p25–26; W8 p27–28; W8 p29 | T8-0016–0018, T8-0021–0029, T8-0062–0064

Short relations and supporting formulas / 短关系式与辅助公式

EN: The first priority is understanding each symbol and keeping units consistent. A=εCL and the short sample relations are useful closed-book memory targets. R=mN is supplementary grating theory used to explain sample Q7; the supplied Week 2 slides do not reliably show that formula. The mock calculation gives it explicitly. Travel-time examples are supporting practice, not an explicit Quiz promise.

中文： 第一优先是理解符号、统一单位。A=εCL与短的样题关系式适合作为闭卷记忆目标。R=mN是解释样题Q7的光栅补充理论，提供的Week2课件没有可靠定位到该式；模拟计算题明确给式。传播时间例题属于辅助训练，并非老师明确保证Quiz会出。

Calibration and LOD / 校准与检测限

Formula / 公式： $m=\Delta y/\Delta C$; $LOD=3\sigma/|m|$

EN: Use a linear region; round up only for a minimum integer count.

中文： 使用线性区；只有最小整数数量才向上取整。

Beer–Lambert and transmission / 吸光度与透射率

Formula / 公式： $A=\varepsilon CL=-\log_{10}T$; $T=I/I_0$

EN: Check mol/L, cm and matching ε units.

中文： 核对mol/L、cm和相匹配的ε单位。

Photon energy / 光子能量

Formula / 公式： $E=hc/\lambda$; $E(eV)\approx1240/\lambda(nm)$

EN: Here c is the vacuum light speed and λ is the vacuum wavelength. Longer wavelength means lower photon energy.

中文： 这里c为真空光速，λ为真空波长。波长更长意味着光子能量更低。

Fluorescence / 荧光

Formula / 公式： $\Phi=k_r\tau$; $k_r=\Phi/\tau$; $\tau=1/(k_r+k_{nr})$

EN: Φ uses absorbed photons; rate units are inverse time.

中文： Φ的分母是吸收光子；速率单位为时间的倒数。

Grating resolution / 光栅分辨率

Formula / 公式： $R=\lambda/\Delta\lambda=mN$

EN: Here m is diffraction order, not a calibration slope.

中文： 这里m是衍射级次，而非校准曲线斜率。

Microscope resolution / 显微分辨率

Formula / 公式： $NA=n\sin\theta$; $d\approx0.61\lambda/NA$

EN: Use the same wavelength and criterion when comparing objectives.

中文： 比较物镜时固定波长及分辨率判据。

Frequency and period / 频率与周期

Formula / 公式： $f=N/\Delta t=1/T$; $\omega=2\pi f$

EN: N counts complete cycles; ms must be converted to s.

中文： N表示完整周期数；ms需要换算成s。

Basic electrical relations / 基础电学关系

Formula / 公式： $V=IR$; $Q=It$; $V_m=V_{inside}-V_{outside}$

EN: Q=It uses constant or time-averaged current. 1 A=1 C/s; 1 V=1 J/C. Distinguish Ω, kΩ and MΩ.

中文： Q=It中的I为恒定电流或时间平均电流。1 A=1 C/s；1 V=1 J/C。区分Ω、kΩ与MΩ。

Differential gain / 差分增益

Formula / 公式： $V_{out}=G(V_1-V_2)$

EN: If G is dimensionless, inputs and output have consistent voltage units.

中文： G无量纲时，输入和输出电压单位必须一致。

Travel time: supporting example / 传播时间：辅助示例

Formula / 公式： Echo: $z=vt/2$; one-way: $z=vt$

EN: Apply only after the question specifies echo versus one-way timing.

中文： 先确认题目给的是回波时间还是单程时间再代入。

EN: A quick calculation workflow: identify the measured variable → write the short relation → convert units → substitute → check whether the answer is continuous or an integer count. A distractor often comes from using the wrong denominator, forgetting the return journey, or reversing a potential difference.

中文： 短计算流程：确认测量量→写短关系式→统一单位→代入→判断结果是连续量还是整数数量。干扰选项常来自分母用错、忘记往返或电位相减方向反了。

Corrections and unresolved sample details / 纠错与样题未决细节

EN: Sample Q10: corrected B=5. Sample Q5: approximately 45 Hz, reference C=40–48 Hz, with limited graph-reading precision. Sample Q9: B is a derived CCD/CMOS reference, while the spoken review hesitates rather than giving a clear final answer. Sample Q12: the intended numerical option is A≈0.55, but the current versus current-density units are inconsistent; do not hide this dimensional issue.

中文： 样题Q10：修正B=5。Q5：约45 Hz，参考C=40–48 Hz，读图精度有限。Q9：B来自CCD/CMOS原理推导；口头回顾有犹豫，没有明确最终字母。Q12：题意数值选项为A≈0.55，但电流与电流密度单位不一致，不能隐去量纲问题。

EN: Source-note corrections: the typed past sheet calls X-rays “non-ionizing”; that is wrong. X-rays are ionizing radiation. The Week 6 p83 inductor relation is printed incorrectly as 2πL/f; the correct ideal reactance is X_L=2πfL. Lifetime is a 1/e decay time, not automatically a half-life; depolarization can make the cell interior positive. These points are corrected rather than copied from notes or ASR.

中文： 源材料纠错：往年打字一页纸把X射线写成“non-ionizing”，错误；X射线属于电离辐射。Week6 p83把电感关系式印为2πL/f，正确理想电抗是X_L=2πfL。Lifetime是1/e衰减时间，不直接等于半衰期；去极化时细胞内可变正。这些内容已纠正，不照抄笔记或字幕识别错误。

EN: PMT correction: the lecture description at T4-0049–0050 wrongly describes tiny mirrors and optical reflection. The actual diagram (W4 p35) labels dynodes: multiplication is by secondary electron emission. Amplification does not remove all noise. See Hamamatsu’s PMT explanation.

中文： PMT纠错：T4-0049–0050把结构说成小镜子与光反射，讲解不准确。原图W4 p35标有dynodes倍增极，实际为二次电子发射倍增；放大不等于消除全部噪声。已核对 Hamamatsu官方PMT说明。

EN: The X-ray “non-ionizing” error also occurs in the original W4 p7, not only a student sheet. Correct it to ionizing; see FDA medical X-ray imaging. W7 p84 also misprints gamma as <25 Hz; learn the band ordering here rather than memorizing this inconsistent boundary.

中文： X射线“non-ionizing”的错误也出现在原课件W4 p7，不只是学生一页纸；应改为电离性，见FDA医学X射线说明。W7 p84还把gamma误写为<25 Hz；这里保留频带顺序，不让你死背不一致的边界。

EN: Further wording safeguards: the glucose/peroxide example is not universal to amperometry; high input impedance is not voltage gain; below-threshold neurons still have a resting potential; ECG leads are not the P/QRS/T waves. These conceptual corrections take precedence over loose oral wording.

中文： 另已防止四种过度概括：葡萄糖/过氧化氢例子不代表全部安培传感；高输入阻抗不等于高电压增益；未达阈值仍有静息电位；心电导联不等于P/QRS/T波。复习采用纠正后的概念，不照抄宽泛口头表述。

Use the materials for revision / 怎么用这套资料

EN: First pass: learn the A cards in English and explain each in Chinese. Second pass: cover the Chinese text and identify the method, measured quantity and relevant short relation. Third pass: take the 25-question paper in English-only mode in 60 minutes; then review every wrong or uncertain item using its source anchors.

中文： 第一遍：读A重点的英文，并用中文解释。第二遍：遮住中文，说出方法、测量量和短关系式。第三遍：切换纯英文模式，60分钟完成25题；再按来源定位复盘每道错题或不确定题。

EN: For final-exam one-page sheets, use only the overlapping concepts for this Quiz. Keep long final calculations for later preparation. The comparison report explains why some frequent final topics were lowered and why explicit classroom Quiz items were raised.

中文： 往年期末一页纸在这次Quiz中主要用于对照重合概念；长期末计算留待之后。对比报告说明哪些期末常见内容降低优先级，哪些课堂明确点名的Quiz内容上调。

EN: Before the mock, scan B01–B32 and say each mechanism in English. After a correct answer, also explain why the distractors are wrong. A high mock score does not establish coverage of all lecture concepts.

中文： 做模拟前扫过B01–B32，并尝试用英文说出各机制。做对后也要解释干扰选项为什么错。模拟卷高分并不代表全部课堂概念都已掌握。